

Dynare + MATLAB Quick Start (Windows)

Use this guide to run a Dynare **.mod** file in MATLAB on Windows.

1. Install MATLAB

Install MATLAB from MathWorks and confirm that it opens correctly.

2. Install Dynare

Download and install Dynare from the official Dynare website:

<https://www.dynare.org/download/>

3. Save the model file

Place your **.mod** file in a simple folder. Example:

```
C:\research\dsti_model\
```

4. Open MATLAB

Start MATLAB normally from Windows.

5. Add Dynare to MATLAB path

Run this in the MATLAB Command Window:

```
addpath C:\dynare\6.0\matlab
```

6. Go to the model folder

Run this in MATLAB, using the folder where your file is saved:

```
cd 'C:\research\dsti_model'
```

7. Check that the file is there

Run:

```
dir
```

8. Run the model

Run the Dynare command. Do not type the **.mod** extension:

```
dynare dj4dsti_mopo_dsti
```

9. If the file includes OSR

OSR means **Optimal Simple Rules**. It asks Dynare to choose policy-rule coefficients that minimize the weighted volatility of selected variables.

Keep the OSR block near the end of the **.mod** file. The important command is:

```
osr(order=1, irf=20);
```

Use **order=1** because OSR without a planner objective only supports first-order approximation.

Example OSR block

```
// =====  
// Optimal policy  
  
w_y=0.5; w_pi=0.5; w_dexc=0.5; w_cr=0.5; w_rl=0.5;  
  
optim_weights;  
y w_y;  
pi w_pi;  
d_exc w_dexc;  
cr w_cr;  
rl w_rl;  
end;  
  
osr_params chi_pi chi_y chi_dexc varepsilon2 varepsilon3;  
  
osr(order=1, irf=20);  
  
v_y = oo_.var(1,1)*10000;  
v_pi = oo_.var(2,2)*10000;  
v_dexc = oo_.var(3,3)*10000;  
v_cr = oo_.var(4,4)*10000;  
v_rl = oo_.var(5,5)*10000;  
  
loss_opt = 10^5*[w_y*v_y + w_pi*v_pi + w_dexc*v_dexc + ...  
                w_cr*v_cr + w_rl*v_rl, ...  
                v_y, v_pi, v_dexc, v_cr, v_rl]';  
  
opt_params = struct2array(oo_.osr.optim_params);
```

10. What Dynare does after you run the file

Reads the file: Dynare reads variables, parameters, equations, shocks, and commands.

Computes the steady state: It calculates the model's long-run equilibrium.

Checks stability: It checks whether the model has a valid dynamic solution.

Solves the model: It computes how variables respond over time.

Runs simulations: It produces impulse response functions and numerical results.

Runs OSR, if included: It searches for the policy-rule coefficients listed in `osr_params`.

Saves results: It stores results in MATLAB objects such as `oo_`, `M_`, and `options_`.